

Proposal, Slitting Machine Retrofitting Project

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Introduction

To automate the metal gang slitting process. The entire manufacturing process was revised and recommendations about adding additional machinery to handle steps that would otherwise be performed by workers manually were suggested. In addition to the physical handling of materials a small prototype of the system was created to test some of the suggested preemptive safety features for the slitting machine and the other machines. Furthermore, the prototype includes a software architecture to enable close to real time monitoring of the physical system.

Problem Statement

Company Inc. currently utilizes a manual process for its industrial slitting machine, which requires human operators to feed and cut raw materials (galvanized steel, aluminum, stainless steel) by hand. This labor-intensive method creates operational inefficiencies and potential safety risks. The objective is to design an automated solution that removes the need for manual intervention during the feeding, cutting, and handling processes.

Objectives

The primary goal is to conceptualize a fully or semi-automated system for the existing slitting machine. Key objectives include:

- Developing an efficient mechanism to manage raw material intake and post-cut material handling.
- Integrating robust safety protocols to eliminate human injury risks.
- Establishing a design that leverages advanced technology while maintaining industrial-grade reliability.

Proposed Solution

The proposed approach focuses on a modular automation design:

- **Automation Logic:** Utilize a PLC to provide the necessary computing power and decision-making capabilities.
- **Material Handling:** Add an automated **conveyor system** to transport raw material into the entry table of the slitting machine, to adjust the orientation of the metal sheets a **motorized lead screw mechanism** is placed before the metal sheet enters the shears of the slitting machine and in parallel a **pneumatic roller** presses the metal sheet down to prevent any bent, bowed or warped sheet from jamming the slitting machine. To load the raw material into the conveyor system a

robot arm will be used and an **automated guided vehicle** will transport the raw and processed material to and from the slitting machine.

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Product Description

[Industrial Sheet Metal Gang Slitting Machine](#) is a High-performance sheet metal coil slitting machine designed for continuous industrial use in HVAC manufacturing, roofing production, and metal fabrication environments.

- Compatible with galvanized steel, aluminum, stainless steel.
- Maximum working width: 55 inches
- Stable performance for 16–30 gauge sheet metal (depending on configuration).



Figure 1

Note the metal sheers are adjustable as shown by the seller:

 Slitting_Machine_Adjustable_Sheers_Video.mp4

Steps Of The Metal Sheet Slitting Process

The steps that a worker would take to handle and process the raw material to and from the slitting machine.

A worker will transport the metal sheets to the slitting machine work area. It will then adjust the slitting machine to the correct width. It will then proceed to place the metal sheet into the slitting machine and push it forward, maintaining a steady speed and ensuring the metal sheet is aligned correctly.

In case of a jam the worker will attempt to pull the metal sheet back and remove any material between the sheers. The cut metal sheet pieces will fall to the back of the slitting machine onto an open container. When the container is full it will be moved to the storage area and replaced by an empty container.

Assumption: We are working with pieces not a metal foil sheet and these metal sheets have the same dimensions.

Proposed Solution

Automating the Slitting Machine Process

What parts of the metal sheet slitting process can be automated and how?



Figure 2: *Autonomous Guided Vehicle to transport the raw and cut metal sheets to and from the slitting machine.*

The transportation process of raw and processed material can be offloaded to autonomous guided vehicles.



Figure 3: *Convoyer Machine to move the metal sheets towards the slitting machine.*

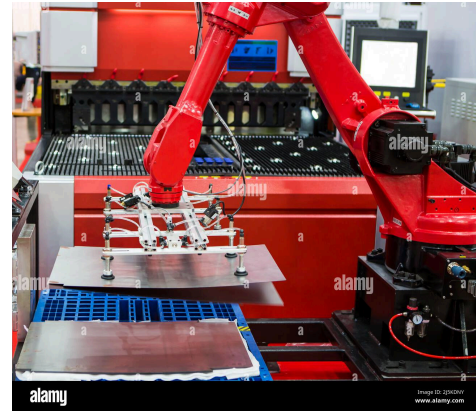


Figure 4: *Robot Arm to move the metal sheets onto the convoyer.*

The lifting and placement of metal sheets into the slitting machine can be performed by a robot arm or a conveyor machine.



Figure 5: *T-slot extrusion for building the overhead frame. Source: DigiKey*



Figure 6:
 *Pneumatic Roller CNC Hold Down on P...*

To handle bent, bowed, or warped sheets without requiring human hands near the arbors, the infeed unit needs active vertical hold-down pressure combined with lateral centering. To achieve

this a pneumatic roller can be added to the existing entry table to press down on the metal sheet evenly, along an overhaul structural frame to support the addition of the other components, and dual direction screw linear actuator mounted to flush across the width of the table to center the metal sheet.

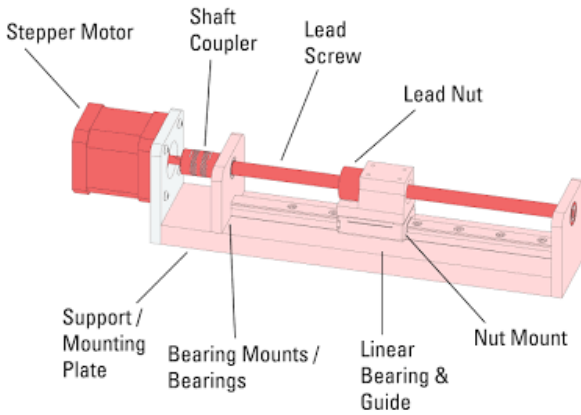


Figure 7: Motorized lead screw mechanism for side guides. Source: Thomson Linear

Risk And Mitigation Strategies



Figure 8: *The preceding image shows how the machine can be set up to accommodate the following safety features.*

Ways to ensure the safety of workers while using the mechanical slitting machine. To create these preemptive safety systems I examined different scenarios and broke the handling of raw material into states; pre,during, and post slitting process.

Infeed & Material Handling Zone (Convoyer/ Robot Arm Input)

The main risk here is material handling, positioning, and sheet metal buckling under automated robot arm movement.

Area Safeguarding(Safety Light Curtains): Mount type 4 safety light curtains on the open access sides of the infeed area. If a robot arm or a worker crosses the perimeter unexpectedly during an active slitting cycle, the light curtain strips the safety relay to stop all upstream and downstream movement immediately.

Interlock Perimeter Fencing : Use a physical wire mesh or steel guards around the gantry robot's operational workspace. Access gates must use high-reliability safety interlock switches (such as trapped key or RFID-coded non-contact switches) integrated into the PLC safety loop.

Point of Operation (The slitting Arbors & Blades)

The slitting arbors create a highly dangerous, high-torque pinch point capable of severe injury.

Mechanically Adjustable Fixed Enclosures: Heavy-duty transparent polycarbonate or sheet-metal tunnel guards must encapsulate the upper and lower arbor shaft. Operators should never be able to reach the blades while material is feeding.

Safety-Interlock Lift-Up Hood: The machine has a hinging hood to manually set the blade configuration or clearing jams, this hood must have an emergency interlock switch. The machine must not run if the hood is lifted.



Figure 9

Zero-Speed/ Delay-to-Unlock Monitoring: Implement a safety dual-channel timer or rotation sensor. Because the arbor shafts have rotation inertia, the safety interlock must remain mechanically locked until the motor has completely decelerated to a zero-speed state.

Buckling & Jam Detection Logic: Monitoring motor current/torque spike on the VFD or using an optical sensor at the discharge table. If the infeed conveyor is moving but no metal breaks the exit sensor within N seconds, immediately trip a controlled stop to prevent severe blade damage or projectile metal debris.

Machine Controls & Emergency Systems

The retrofitted system requires hardwired control overrides that people can interact with independently of standard software logic.

Emergency Strop(E-Stop) Circuitry: Install prominent, mushroom-head E-stop buttons at the control box, the entry table and the exit table, These must be wired directly to a dedicated safety relay or safety PLC, using dual-channel, normally closed (NC) fail-safe circuit configuration.

Emergency Grab Wires(Cable Pulls): Run safety grab-wire cables along the entire length of the conveyor and slitting line. An operator or by stander can pull the cable anywhere along the line to instantly kill power to the variable frequency drive (VFD) and primary motor.

Two-Hand Control / Jog Mode: For maintenance procedures, clearing jams, or threading the initial coil roll, require a two-hand control panel or a strictly regulated “jog” speed mode that bypasses full automation only under highly restricted human control.

Process State Machine

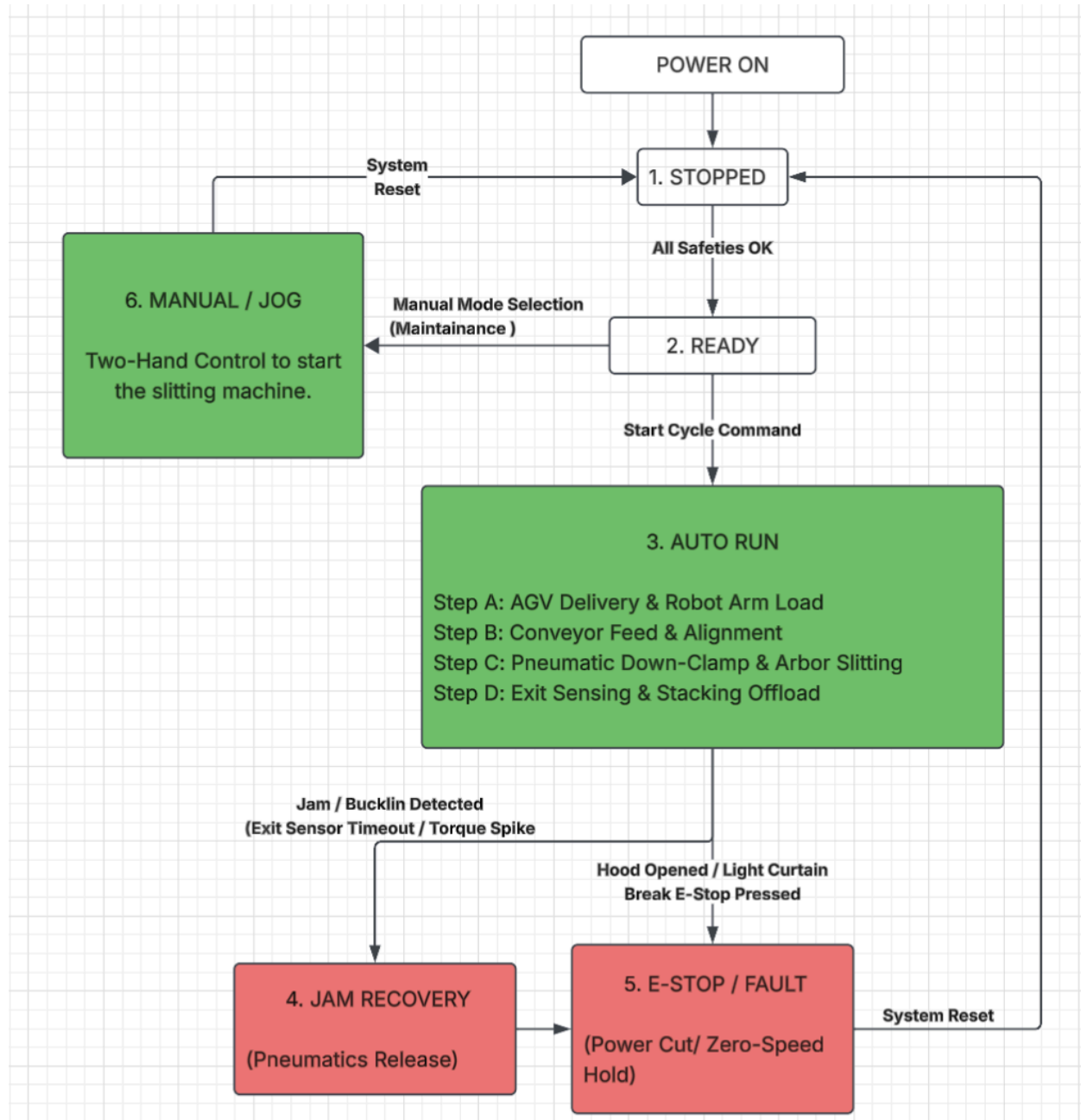
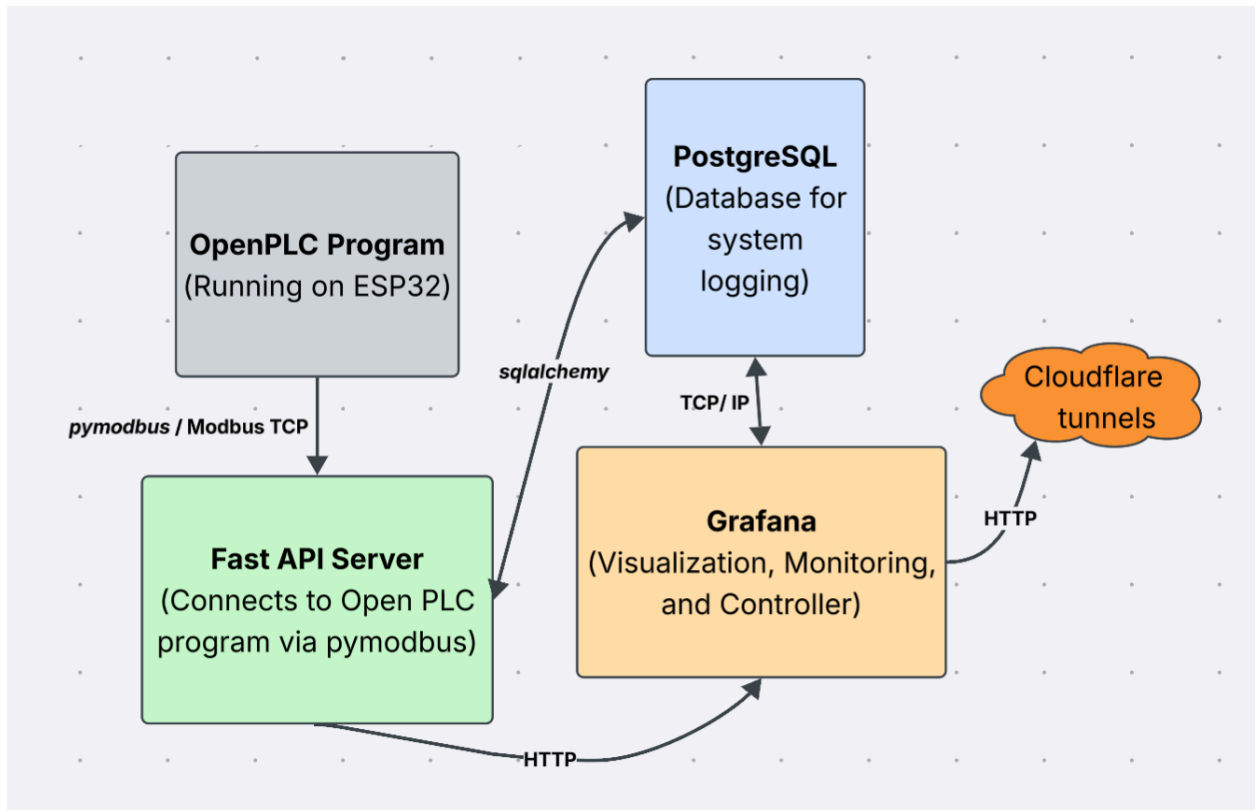


Figure 10

OpenPLC firmware & Cloud Integration Architecture



Physical Prototype

Running Open PLC on a ESP32 DevKitV1 Micro-controller, Fast API, PostgreSQL, Grafana with Cloudflare tunneling for access outside the local area network.

Dashboard Link

Code [Link](#)

Prototype Image

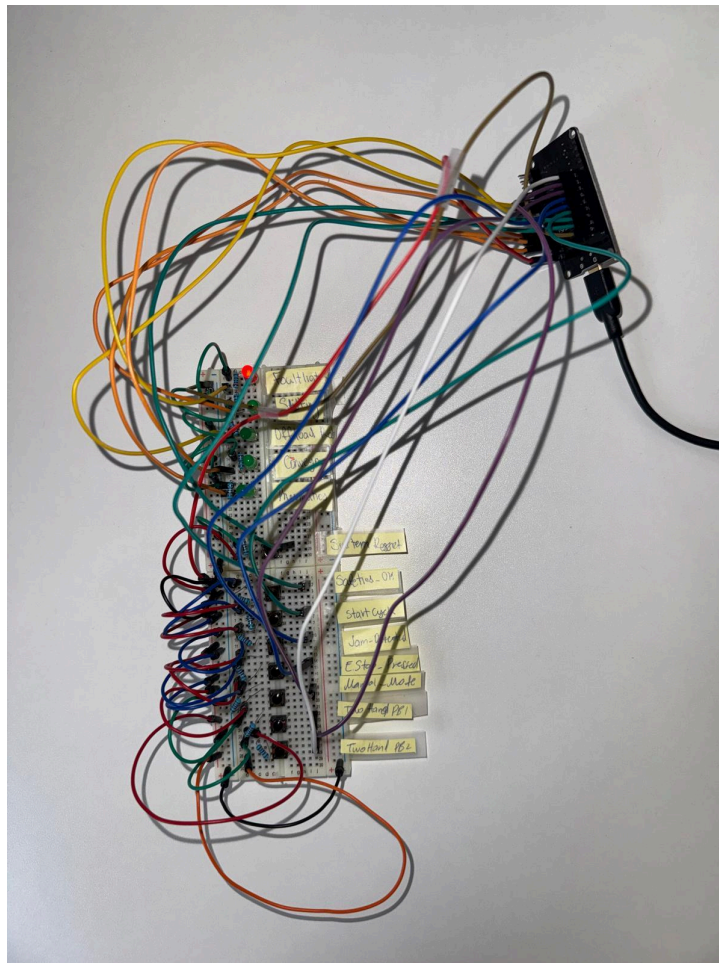


Figure 11

The buttons set the Open PLC program to different states such as Manual Mode, E-Stop Pressed, System Reset, Safeties_OK, Start Cycle, Jam Detected. The LEDs represent physical machinery that's part of the automation system such as conveyors, pneumatic press rollers.

References

- 1) <https://edge.autonomylogic.com/docs/openplc-editor/overview>
- 2)  How to Use OpenPLC (Free PLC Software) | ESP32 PLC
- 3) **OSHA 29 CFR 1910.212** standards